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(*b*) Write a note on text editor. What are important editing functions? 7

(*c*) What are the constituents of an operating system? Explain any two of them. 7

(*d*) Differentiate between assembler, cross assembler, line by line interpretor and a compiler. 7

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324614(24)

BE (6th Semester)
Examination, April - May, 2016

(Old Scheme)

Microprocessors

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Time Allowed : 3 hours *Maximum Marks :* 80

Minimum Pass Marks : 28

Note : (*i*) Part (*a*) of each unit is compulsory. Attempt any **two** parts from (*b*), (*c*) and (*d*).

(*ii*) The figures in the right-hand margin indicate marks.

1. (*a*) During i/o mapped i/o mode, when an i/o is addressed, its address comes on A_{15}–A_8 lines. What are the contents of AD_7–AD_0 lines, during this time? 2

(*b*) Draw funtional organisation diagram of 8085. Label the diagram and explain its general purpose registers. 7

(*c*) Explain all conditional flags of 8085. 7

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(*d*) Explain LHLD addr; STAX rp; INX rp; DAD rp; DAA; RRC and SPHL instructions of 8085. 7

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2. (*a*) Make a table and compare Static and Dynamic RAMs on at least 4 points. 2

(*b*) What are different electrical signals on a typical memory chip? Draw and explain them. 7

(*c*) How is 6264 organised? Draw, label and explain read cycle of 6264 static RAM. 7

(*d*) Explain refreshing of DRAMs including $\overline{RAS}$ only refresh. Hidden refresh, burst and distributed refresh. Draw figures wherever needed and neatly lable them. 7

3. (*a*) How can bus contention be avoided using chip enable and output enable? What is constraint with reference to t_{CE} and t_{DF}? 2

(*b*) Use 74LS138 to generate signals to chip enable of memory and i/o for address map as below : 7

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Address From	to	Device to be Selected
0000	1FFF	2764
2000	3FFF	6264 # 0
4000	5FFF	6264 # 1
6000	7FFF	6264 # 2
8000	9FFF	6264 # 3
A000	BFFF	6264 # 4
C000	DFFF	6264 # 5
E000	E000	i/o # 1
E001	E001	i/o # 2

Draw figure, label and explain it. Do i/os have unique address.

(*c*) What is polling, when is it needed, what is software and hardware polling? 7

(*d*) Write a note on enabling, disabling and masking of interrupts. Explain use of RIM and SIM instructions, with ACC contents after and before execution of RIM and SIM instructions. 7

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4. (*a*) Explain working of 8212 with a neat schematic diagram, label the diagram. 2

(*b*) How can we cascade 8259? Explain with figure to interface 15 i/o devices connected to 8259 such that each device has a unique ISS address. 7

(*c*) Explain $\overline{CS}$; $C/\overline{D}$; $\overline{WR}$; $\overline{RD}$; RESET; CLR signals of 8251. Also explain the control register, status register and data buffer of 8251. 7

(*d*) Explain working of AD557 with the help of figures showing ADSS7 signals and its timings. 7

5. (*a*) What is a system monitor and how it helps a programmer? 2

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